

SauceDemo 3-Layer Architecture

LAYER 1 — BDD / CUCUMBER

Feature Files

.feature (Gherkin)
6 features, 20 scenarios
Tags: @smoke
@functional

Runners

*Runner.java
@RunWith(Cucumber)
JUnit4, per-feature

Step Definitions

*Steps.java
@Given @When
@Then
Thin — delegates to services

Hooks

Hooks.java
@Before
Sets
Screenshots per scenario

Config

.properties

login_validation order_pdf_receipt product_sort
product_image_integrity session_management visual_regression

▼ calls

LAYER 2 — SERVICES (Business Logic)

LoginService

login, error checks
LoginService.java

CheckoutService

cart → checkout → PDF
CheckoutService.java

ProductCatalogService

sort, prices, images
ProductCatalogService.java

SessionService

reset, logout, back
SessionService.java

Services orchestrate page objects — no Playwright calls, no assertions. Steps call services; services call pages.
Returns DTOs (ReceiptSummary, LineItem) so steps can assert on typed data, not raw strings.

▼ calls

LAYER 3 — PAGE OBJECTS (UI Interaction)

LoginPage

data-test locators
LoginPage.java

InventoryPage

products, sort, cart
InventoryPage.java

CartPage

checkout()
CartPage.java

CheckoutPages

StepOne → Two → Done
Checkout*.java

NavigationMenu

reset, logout
NavigationMenu.java

BasePage

abstract parent
BasePage.java

Each page owns its Playwright locators (data-test attributes). Returns typed data, never raw DOM elements.
Checkout pages chain: fillInfoAndContinue() → CheckoutStepTwoPage, finish() → CheckoutCompletePage

DTOs (Data Objects)

LoginCredentials

user, pass

CheckoutInfo

name, zip

LinItem

name, BigDecimal

ReceiptSummary

items, tax, total

DTOs carry typed data between layers.
Steps assert on DTOs, not strings.

UTILITIES

DriverManager

ThreadLocal Playwright

PdfReceiptParser

PDFBox extraction

ImageComparator

pixel-diff visual

ScreenshotHelper

on-demand capture,
embed in Cucumber reports

EXAMPLE FLOW: PDF Receipt Verification

Feature File

→

Steps

→

CheckoutSvc

→

Pages (POM)

→

PdfParser

→

ReceiptSummary

(DTO)

Step asserts: ReceiptSummary.total == expected \$43.18, tax == \$3.20, line items match, PDF total == on-screen total